V U _ S U R V E Y O R S

NASA SPACE APPS CHALLENGE 2026

FireFusion — System Design Document

Harmonization of MODIS and VIIRS Hot Spots

Confidence-Aware, Sensor-Harmonized Wildfire Intelligence Platform

Detail	Information
Team	VU_Surveyors
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01 EXECUTIVE SUMMARY

Challenge 9 — Harmonization of MODIS and VIIRS Hot Spots — asks teams to build a web application that merges two satellite fire-detection datasets (NASA MODIS and VIIRS) into a single, trustworthy "burning activity calendar." The core difficulty is not simply overlaying two datasets on a map; it is creating a scientifically defensible cross-sensor harmonization layer that reconciles fundamentally different sensor characteristics.

This is a strong pick for a developer-led team. NASA FIRMS data is openly accessible, the work is primarily data engineering plus statistics, and a compelling demo — a raw-versus-harmonised toggle over a burning-activity calendar that visibly removes a sensor artefact — is achievable within 48 hours.

WHY NOW

Suomi-NPP data delivery ends 1 November 2026 and the MODIS instruments on Terra and Aqua are being retired over 2026–27. Continuity between the two sensor families is now a practical problem, not an academic one.

Core Recommendation

Treat harmonisation quality as the product. The judges' stated definition of success is an accurate, time-based map of when and where fires occurred that reconciles MODIS and VIIRS. Analyst briefs, priority scores and LLM text are nice extras, but they only score if the reconciled series is defensible and its validation is visible on screen.

Four Critical Corrections to Existing Plans

- **Do not filter on one confidence number. MODIS confidence is numeric (0–100) while VIIRS is categorical (low/nominal/high). Map both to a common three-class scale first.**
- **Grid-collapsing alone will not remove the sensor artefact. VIIRS also detects smaller fires MODIS cannot see. A second step — detectability matching plus a fitted sensor factor — is needed.**
- **Use local dates for the calendar. FIRMS timestamps are UTC; night passes over Bangladesh fall on the previous UTC date. Store UTC, build the calendar on Asia/Dhaka date.**
- **Verify the killer-demo claim before promising it. "Counts triple at the transition" is illustrative — measure it on real data and let the measured number go into the pitch.**

02 PROBLEM ANALYSIS & SCIENTIFIC FOUNDATION

2.1 The Harmonization Problem

A harmonisation design is only as good as the list of differences it handles. MODIS and VIIRS are not equivalent sensors. They fly on different platforms, have different spatial resolutions, different detection sensitivities, and use different confidence metrics. When you naïvely stitch them together, you create a false discontinuity in the fire record.

The build guide frames the central issue: a single burning area may produce one MODIS detection but 5–7 VIIRS detections due to the finer spatial resolution. If you naïvely count detections, you could incorrectly conclude that fire activity increased sevenfold, when what actually happened is that the higher-resolution sensor resolved one burning area into multiple detections.

KEY Harmonization, rather than simple merging, is the core scientific problem. The goal is to make the combined record tell a consistent story across the sensor transition.

2.2 Sensor Comparison Matrix

Dimension	MODIS (Terra, Aqua)	VIIRS 375 m (S-NPP, NOAA-20/21)
Nominal pixel	1 km at nadir, grows toward scan edge	375 m at nadir (I-band), smaller growth
Record start (FIRMS)	Terra Nov 2000, Aqua Jul 2002	S-NPP 2012; NOAA-20/21 later
Overpass (local solar)	Terra ~10:30/22:30; Aqua ~13:30/01:30	~13:30/01:30 (afternoon orbit)
Confidence	Numeric 0–100	Categorical: low / nominal / high
Intensity fields	brightness (band 21), bright_t31, FRP	bright_ti4, bright_ti5, FRP
Footprint fields	scan, track (km)	scan, track (km) — much smaller
Sensitivity	Misses many small/cool fires	Detects smaller, cooler fires
Timestamp	acq_date + acq_time in UTC	Same format, UTC
Continuity	Being retired 2026–27	S-NPP ends 1 Nov 2026; NOAA-20/21 continue

2.3 Two Effects That Must Be Separated

SCIENCE

Two different effects are mixed in a raw count series and must be separated:

(a) Resolution / duplication: One real fire produces several VIIRS pixels but one or two MODIS pixels. Fix: cluster detections into events.

(b) Sensitivity: VIIRS detects fires that MODIS cannot. Clustering does not remove this. Fix: detectability matching + estimate a residual sensor factor on the overlap period, with uncertainty.

Observation vs. Event — The Key Distinction

Your system should distinguish two concepts clearly:

Concept	What It Is	Rule
Observation	A raw satellite detection from FIRMS	Evidence — never destroy it; keep the original record intact
Harmonized Fire Event	Your derived representation of burning activity	Interpretation — cluster observations, apply calibration, carry uncertainty

03 PRODUCT CONCEPT & SCOPE

3.1 Product Vision — FireFusion

FireFusion is a web platform that ingests MODIS and VIIRS hotspot records for a chosen area, produces a harmonised, source-preserving burning-activity series, and shows it as a calendar heatmap and map with a raw-versus-harmonised toggle, cluster-level evidence and an explicit uncertainty display.

The system answers five fundamental questions:

20. Where is fire activity happening?
21. When is it happening?
22. How much of the apparent increase is real versus sensor-related?
23. Are multiple sensors observing the same likely burning event?
24. Is today's activity unusual compared with the historical baseline for that place and season?

PRODUCT FireFusion is sensor-aware wildfire intelligence — not wildfire prediction AI. It harmonizes observations, validates the harmonization, detects anomalies, and explains evidence. It never claims to predict fire spread or danger.

3.2 Functional Requirements

ID	Requirement	Priority
F1	Ingest FIRMS MODIS and VIIRS records for a configurable bounding box and date range; cache to Parquet	Must
F2	Canonical schema: sensor family, platform, UTC timestamp, local date, confidence class, FRP, footprint	Must
F3	Cluster detections into fire events (DBSCAN haversine); keep source sensor on every record	Must
F4	Produce raw and harmonised daily series for any selected area	Must
F5	Calendar heatmap (year × day) with raw / harmonised toggle	Must
F6	Overlap-period validation shown in the interface (correlation and ratio, raw vs harmonised)	Must

F7	Cluster evidence panel (which sensors, times, FRP, confidence)	Should
F8	Seasonal baseline and anomaly percentile for a chosen date and area	Should
F9	Uncertainty bands (bootstrap intervals) on harmonised series	Should
F10	Template-based analyst brief with source citations (LLM optional)	Could
F11	Methods and limitations page ("what this means / does not mean")	Must
F12	Confidence-aware event scoring with transparent, configurable weights	Should

3.3 Non-Functional Requirements

- Offline-first: the demo reads only from local cache; OFFLINE=1 forces fixture mode; a recorded fallback video exists
- Reproducible: one command rebuilds every derived file from raw downloads; fixed random seeds; pinned dependencies
- Transparent: every number on screen traces to a dataset ID, product name, date range and parameter set
- Fast: any API call for the pilot region returns in well under one second from pre-aggregated tables
- Compliant: open-source licence (Apache-2.0), AI-use disclosure in docs/AI_USE.md, no likeness of anyone under 18 in video
- Testable: deterministic synthetic test cases for every compute function; fixed seeds for any stochastic step

3.4 Pilot Scope & Non-Goals

Scope

- Area: Bangladesh-centred bounding box (lon 88–93, lat 20–27); configurable for wider South Asia
- Time range: Full overlap period (2012–present) for model; MODIS-only years back to 2000 for long calendar
- Quality filter: nominal-or-high confidence class after mapping; low class kept in raw layer
- Optional secondary region: Australian bushfire zone or Amazon basin for contrast

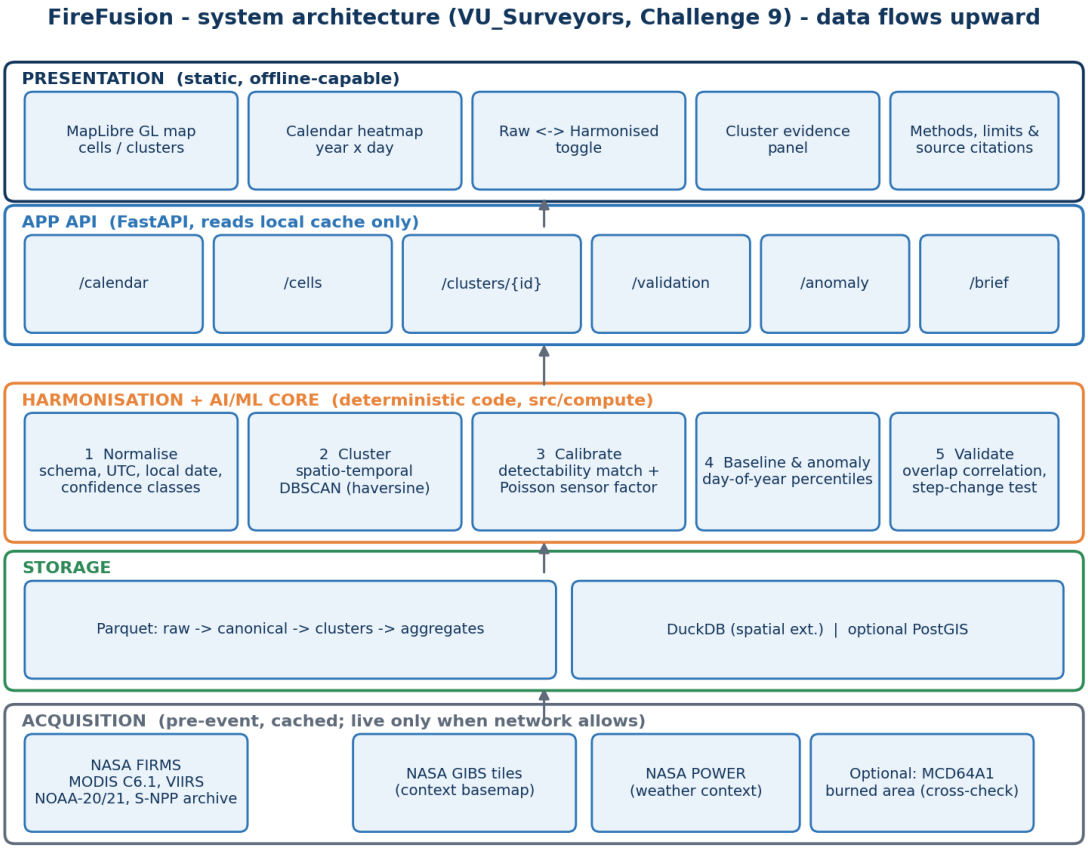
Non-Goals

- Fire-perimeter mapping or fire-spread simulation
- Evacuation guidance or real-time emergency alerts
- Weather prediction or wind modelling
- Mobile app (web is sufficient)
- Complex microservices or Kubernetes orchestration
- LLM-based fire prediction (scientifically indefensible)

04 SYSTEM ARCHITECTURE

4.1 High-Level Architecture Diagram

The design follows the guide's layer rule: the frontend never calls NASA directly, the API never calls NASA during a demo, and acquisition runs before the event. Storage is a chain of Parquet files queried with DuckDB, so the whole system runs on a laptop without a database server.



4.2 Component Responsibilities

Layer	Component	Responsibility	Technology
Data	Acquisition	Bulk FIRMS archive download; area-API for latest days; GIBS tiles; NASA POWER; retry + caching + fixture fallback	Python, requests, pandas
Data	Normalisation	Map product-specific columns to canonical schema; parse UTC; derive local date; map confidence classes; flag static sources	pandas / DuckDB, pydantic
Compute	Clustering	Group same-day detections across sensors into fire events; store member IDs for evidence panel	scikit-learn DBSCAN (haversine)

Compute	Calibration	Detectability matching; Poisson GLM sensor factor estimation on overlap period with confidence intervals	statsmodels, scipy, LightGBM
Compute	Analytics	Daily series (raw + harmonised), day-of-year baseline, anomaly percentile, aggregates	DuckDB SQL, numpy
Compute	Validation	Overlap correlation, ratio, step-change test, sensitivity analysis, unit tests	pytest, scipy
API	FastAPI Server	Read-only endpoints: calendar, cells, clusters, validation, anomaly, brief, meta	FastAPI, uvicorn, orjson
Frontend	Web Client	Map, calendar heatmap, toggle, evidence panel, methods panel (teammate responsibility)	MapLibre GL, D3 / Observable Plot
Optional	Explainer Agent	Turns structured numbers into a brief with citations; never computes; template fallback	Claude API, Jinja2 templates

4.3 Data Flow Pipeline

Data flows through five deterministic stages, each producing a Parquet artifact that can be inspected and validated independently:

Stage	Input	Output	Key Operation
1. Acquire	NASA FIRMS API / archive	raw/*.parquet (unmodified FIRMS columns)	Download, validate, cache
2. Normalise	raw/*.parquet	canonical/*.parquet	Schema mapping, UTC→local, confidence class
3. Cluster	canonical/*.parquet	clusters/*.parquet + detections with cluster_id	DBSCAN per local_date (haversine)
4. Calibrate	clusters/*.parquet (overlap period)	calibration.json (sensor factor + intervals)	Poisson GLM; detectability matching
5. Aggregate	clusters + calibration	agg_daily.parquet, agg_cells.parquet	Pre-compute daily counts per area + per cell

4.4 Offline-First Design

This is not optional for the project. The guide explicitly recommends pre-caching all data and having every live request fall back to cached/fixture data.

Mode	Behaviour
OFFLINE=0 (default)	API reads from pre-computed Parquet cache; acquisition can refresh recent days if network allows
OFFLINE=1 (demo)	API reads only from fixtures in cache/demo_fixtures/; no outbound network calls at all
Fallback video	A pre-recorded screen recording exists as last resort if the live demo fails

TIP During the hackathon demo, OFFLINE=1 means no network dependency. If the demo wifi fails, your product still works. This is extremely valuable.

05 DATA MODEL & SCHEMA DESIGN

5.1 Entity-Relationship Overview

The data model has four main entities, each stored as Parquet files queryable via DuckDB. The ER relationships are: Detection → belongs to → Cluster; Cluster → aggregated into → Daily Aggregate; all linked to Metadata/Provenance records.

Entity	Granularity	Key Fields	Relations
Detection	One satellite pixel	det_id, sensor_family, platform, ts_utc, lat, lon, frp, conf_class	Many-to-one → Cluster
Cluster (Fire Event)	One spatial-temporal group per day	cluster_id, local_date, centroid, n_modis, n_viirs, corroborated	One-to-many → Detections
Daily Aggregate	One row per (area, date, mode)	date, area_id, raw_count, harmonised_count, interval_lo, interval_hi	Derived from Clusters
Provenance	One per dataset/pipeline run	run_id, datasets[], parameters, git_sha, timestamp	Links to all derived tables

5.2 Detection (Canonical) Table

One canonical table for all detections. The original FIRMS columns are preserved in the raw layer so output can always be traced back to the source record.

Field	Type	Notes
det_id	string	Stable hash of product, lat, lon, timestamp — globally unique
sensor_family	enum	MODIS or VIIRS (use this for corroboration, not platform)
platform	enum	Terra, Aqua, S-NPP, NOAA-20, NOAA-21
product	string	FIRMS product name and collection, for citation
ts_utc	timestamp	From acq_date + acq_time (UTC)
local_date	date	ts_utc converted to Asia/Dhaka; used for calendar
lat	float	WGS84 latitude as delivered
lon	float	WGS84 longitude as delivered
footprint_km2	float	scan × track (pixel area)
frp_mw	float	Fire Radiative Power; cross-sensor comparability is limited
conf_class	int (0,1,2)	low/nominal/high after mapping; raw confidence in raw layer
daynight	enum	Day or night flag

static_flag	bool	Known persistent thermal source (e.g., industrial sites)
cluster_id	string	Assigned by clustering stage; nullable before clustering

5.3 Cluster (Fire Event) Table

Field	Type	Notes
cluster_id	string	Unique ID: {local_date}_{cell}_{seq}
local_date	date	The day this fire event was observed
centroid_lat	float	Weighted centroid of member detections
centroid_lon	float	Weighted centroid of member detections
n_modis	int	Number of MODIS detections in this cluster
n_viirs	int	Number of VIIRS detections in this cluster
n_platforms	int	Distinct platforms that saw this event
corroborated	bool	True if both MODIS and VIIRS detected it
first_ts_utc	timestamp	Earliest detection timestamp
last_ts_utc	timestamp	Latest detection timestamp
max_frp	float	Maximum FRP across all member detections
median_frp	float	Median FRP across all member detections
best_conf_class	int	Highest confidence class among members
event_score	float	Composite confidence-aware score (Section 7.3)

5.4 Aggregate Tables

Pre-computed to keep API responses fast. Two aggregate tables serve the calendar and map views:

- agg_daily: one row per (area, local_date, mode). Columns: raw_event_count, harmonised_event_count, interval_lo, interval_hi, n_corroborated, mean_frp, max_frp
- agg_cells: one row per (cell_id, local_date, mode). Columns: centroid_lat, centroid_lon, event_count, corroborated_pct, max_frp, best_conf_class

5.5 Metadata & Provenance Table

Every result carries provenance. This is a first-class feature, not an afterthought.

Field	Type	Notes
run_id	string	UUID for this pipeline execution
git_sha	string	Commit hash of the code that produced this output
datasets	array[string]	List of FIRMS product names and date ranges used
parameters	json	Cluster radius, confidence thresholds, overlap window, etc.
timestamp	datetime	When this pipeline run completed

row_counts	json	Count of detections, clusters, aggregates produced

06 BACKEND API DESIGN (FastAPI)

6.1 API Endpoint Specification

All endpoints are read-only (GET). The API reads only from pre-computed Parquet/DuckDB cache — no live NASA calls during the demo. All responses include a provenance object.

Endpoint	Parameters	Returns
GET /calendar	bbox, start, end, mode=raw|harmonised	Daily values for calendar heatmap with interval bounds in harmonised mode
GET /cells	date, bbox, mode	Grid/H3 cells with event counts for the map layer
GET /clusters/{id}	—	Members, sensors, times, FRP, confidence (evidence panel)
GET /clusters/search	bbox, date, min_score	List of clusters matching spatial-temporal query
GET /validation	bbox	Overlap-period correlations, ratio (raw vs harmonised), step-change test, parameter set
GET /anomaly	bbox, date	Percentile against day-of-year baseline, baseline window, sample size
GET /baseline	bbox, doy	Day-of-year percentile distribution (p10, p50, p90, p95)
GET /brief	bbox, date	Structured numbers object + generated/templated text with source citations
GET /meta	—	Datasets, product names, date ranges, limitations, licence (feeds methods panel)
GET /health	—	Service status, cache freshness, data coverage dates

6.2 Request/Response Schemas

Example: GET /calendar Response

```
{
  "bbox": [88.0, 20.0, 93.0, 27.0],
  "start": "2003-01-01", "end": "2026-09-01",
  "mode": "harmonised",
  "data": [
    { "date": "2003-01-01", "count": 12, "lo": 10, "hi": 15 },
    { "date": "2003-01-02", "count": 8,  "lo": 6,  "hi": 11 },
    ...
  ],
  "provenance": {
    "datasets": ["MODIS_C6.1", "VIIRS_SNPP_NRT"],
    "parameters": { "cluster_eps_km": 1.5, "conf_filter": ">=nominal" },
    "run_id": "abc123", "git_sha": "def456"
  }
}
```

Example: GET /clusters/{id} Response

```
{
  "cluster_id": "2024-03-15_h3_8823_001",
  "local_date": "2024-03-15",
  "centroid": [24.123, 90.456],
  "corroborated": true,
  "event_score": 0.87,
  "members": [
    { "det_id": "...", "sensor": "MODIS", "platform": "Aqua",
      "frp": 45.2, "conf_class": 2, "ts_utc": "2024-03-15T07:30Z" },
    { "det_id": "...", "sensor": "VIIRS", "platform": "NOAA-20",
      "frp": 38.1, "conf_class": 2, "ts_utc": "2024-03-15T07:45Z" }
  ],
  "summary": { "n_modis": 2, "n_viirs": 5, "max_frp": 52.3 }
}
```

6.3 Error Handling & Middleware

- All responses wrapped in a standard envelope: { status, data, error, provenance }
- 400 for invalid bbox/date; 404 for unknown cluster_id; 503 if cache is stale beyond threshold
- CORS enabled for the static frontend origin
- Request logging with structured JSON (timestamp, endpoint, params, duration_ms)
- Rate limiting not needed for hackathon; placeholder for production

07 HARMONIZATION ENGINE & AI/ML PIPELINE

RULE Deterministic code does the science. Models estimate and explain, and every estimate is validated. The ML components are small, testable and have clear fallbacks, so a failure in one never blocks the demo.

7.1 Stage 1: Normalisation

Convert all raw FIRMS data to a common canonical schema. This is where sensor-specific quirks are resolved:

Step	What It Does	Implementation Detail
Parse timestamp	Combine acq_date + acq_time into a proper UTC timestamp	pd.to_datetime with format='%Y-%m-%d %H%M', utc=True
Derive local date	Convert UTC to Asia/Dhaka timezone for calendar display	ts.dt.tz_convert('Asia/Dhaka').dt.date
Map confidence	MODIS 0–100 → three classes; VIIRS l/n/h → same classes	MODIS: 0–29→low, 30–79→nominal, 80–100→high (verify in FIRMS docs)
Compute footprint	Calculate pixel area from scan × track	footprint_km2 = scan * track
Tag sensor family	Assign MODIS or VIIRS based on product name	Use sensor_family column, not platform
Generate det_id	Stable hash for deduplication	SHA-256 of (product, lat, lon, ts_utc)

Normalisation Code

```
CONF_VIIRS = {"l": 0, "n": 1, "h": 2}  # low / nominal / high

def to_canonical(df, product):
    is_modis = product.startswith("MODIS")
    ts = pd.to_datetime(df.acq_date.astype(str) + " " +
                        df.acq_time.astype(str).str.zfill(4),
                        format="%Y-%m-%d %H%M", utc=True)
    if is_modis:
        conf = pd.cut(pd.to_numeric(df.confidence),
                      [-1, 29, 79, 100],
                      labels=[0, 1, 2]).astype("Int8")
    else:
        conf = df.confidence.map(CONF_VIIRS).astype("Int8")
    return pd.DataFrame({
        "sensor_family": "MODIS" if is_modis else "VIIRS",
        "platform": df.satellite, "product": product,
        "ts_utc": ts,
        "local_date": ts.dt.tz_convert("Asia/Dhaka").dt.date,
        "lat": df.latitude, "lon": df.longitude,
        "footprint_km2": df.scan * df.track,
        "frp_mw": df.frp,
        "conf_class": conf, "daynight": df.daynight
    })
```

7.2 Stage 2: Spatio-Temporal Clustering (Fire Events)

Baseline method: the guide's grid collapse (simple, explainable, good fallback). Main method: DBSCAN with haversine metric run per local date across both sensor families.

Parameter	Default	Rationale
eps_km	1.5	Reflects geolocation uncertainty of MODIS (coarser sensor); start 1–2 km
min_samples	1	Every detection matters; no noise rejection at this stage
metric	haversine	Proper great-circle distance on the sphere
grouping key	local_date	One clustering pass per calendar day (Asia/Dhaka time)

Clustering Code

```
from sklearn.cluster import DBSCAN
EARTH_R_KM = 6371.0088

def cluster_day(g: pd.DataFrame, eps_km: float = 1.5) -> np.ndarray:
    X = np.radians(g[["lat", "lon"]].to_numpy())
    return DBSCAN(
        eps=eps_km / EARTH_R_KM, min_samples=1,
        metric="haversine", algorithm="ball_tree"
    ).fit_predict(X)

# Run per local_date:
# labels -> cluster_id = f"{local_date}_{label}"
# Aggregate: n_modis, n_viirs, corroborated, max_frp, best_conf
```

Sensitivity analysis: run clustering at eps_km = 0.75, 1.0, 1.5, 2.0, 3.0 km and report a sensitivity table. Choose the value that maximises overlap agreement without merging distant fires.

7.3 Stage 3: Calibration (The ML Core)

After clustering, the remaining discrepancy is sensitivity — VIIRS sees fires MODIS cannot. Three layers, from simple to ambitious, shipped in order:

Layer	Method	Output & Validation
A. Detectability Matching (MVP)	Restrict VIIRS clusters to those a MODIS-class sensor could plausibly detect: aggregate VIIRS detections to a MODIS-like footprint and apply an FRP threshold chosen on the overlap period to maximise agreement.	A "MODIS-equivalent VIIRS" series. Choose threshold by leave-one-year-out cross-validation; report threshold and agreement metric.
B. Sensor Factor (Should)	Poisson (or quasi-Poisson) GLM on overlap-period daily event counts per district: n ~ sensor + month + district, with robust standard errors (HC1).	A multiplicative factor with confidence interval. Apply to convert between sensors; carry interval into plots. Season-specific version if ratio varies by month.
C. Learned Co-Detection (Stretch)	LightGBM classifier predicting whether a MODIS event is also seen by VIIRS, from FRP, brightness, conf_class, footprint, day/night, local density, month. Calibrated probabilities (isotonic).	Detection-probability surface. Evaluate with spatial-block cross-validation, AUC, calibration curve. Weight events by 1/p. Only ship if it beats Layer B.

Sensor Factor Estimation Code (Layer B)

```
import statsmodels.api as sm, statsmodels.formula.api as smf

# overlap_df: one row per (district, local_date, sensor_family), n_events
m = smf.glm(
    "n_events ~ C(sensor_family) + C(month) + C(district)",
    data=overlap_df,
    family=sm.families.Poisson()
).fit(cov_type="HC1")

ratio = np.exp(m.params["C(sensor_family)[T.VIIRS]"])
lo, hi = np.exp(m.conf_int().loc["C(sensor_family)[T.VIIRS]"])
# Divide VIIRS-era counts by ratio for MODIS-equivalent;
# carry [lo, hi] into plotted uncertainty band
```

Confidence-Aware Event Score

Instead of just counting events, create a composite score with transparent, configurable components:

Component	Weight (default)	Source
Confidence class	0.30	Best confidence among member detections (mapped to 0–1)
Corroboration	0.25	Binary: both sensor families present? (0 or 1)
FRP strength	0.20	Log-scaled max FRP, normalised to [0, 1]
Temporal spread	0.15	Hours between first and last detection (persistence signal)
Platform diversity	0.10	Distinct platforms / max possible platforms

NOTE Do not arbitrarily invent scientific coefficients. Make weights configurable and show all components to the user. This gives transparent harmonization instead of black-box fire prediction.

7.4 Stage 4: Seasonal Baseline & Anomaly Detection

- Baseline: for a chosen area and day-of-year, collect harmonised daily counts within a ±7-day window across baseline years; store selected percentiles (p10, p50, p90, p95)
- Anomaly: empirical percentile of the observed value in that distribution, with window and sample size displayed. Use mid-rank percentile to handle ties (many days have zero fires)
- Optional: STL decomposition or Poisson tail probability for trend-adjusted view. Keep empirical percentile as headline because it is easy to explain
- Language rule: say "unusual for this place and time of year" — never "will spread" or "danger"

7.5 Stage 5: Validation Framework

This is the evidence judges will look for. Validation is not a nice-to-have — it is the core of the Validity score.

Test	What It Shows	Success Criterion
Overlap correlation	On the period both sensors observed, compute correlation of raw MODIS vs raw VIIRS, then harmonised vs harmonised	Harmonised correlation is higher than raw
Overlap ratio	Mean daily count ratio between sensors, raw vs harmonised	Harmonised ratio is closer to 1.0

Step-change test	Segmented regression or CUSUM at sensor handover on stitched long record	Raw shows significant shift; harmonised does not (or shift is within interval)
Parameter sensitivity	Results across cluster radius, confidence filter, cell size	Result is stable across reasonable parameter choices
Bootstrap intervals	Block bootstrap over days/weeks for ratio and harmonised series	Intervals are narrow enough to be useful; reported honestly
Independent cross-check	Compare harmonised seasonal shape with MCD64A1 burned-area timing	Timing agreement (not proof, but supportive evidence)
Unit tests	Synthetic cases: fire seen by both sensors, VIIRS-only small fire, boundary cases	All deterministic tests pass

RULE

Honesty guardrails (also good for the Validity score):
- State on screen that hotspots are detections, not fire perimeters
- Show that MODIS and VIIRS confidence values are mapped to classes, not 1:1
- Show the interval, not only the point estimate
- If the sensor factor differs by season or land type, say so

7.6 LLM/Agent Integration

The language model is optional and confined to explanation. Architecture:

Rule	Description
Input	A JSON object with area, dates, raw + harmonised counts, interval, percentile, validation numbers, dataset names
Output	A short brief that may only quote numbers present in that input JSON
Fallback	A Jinja2 template produces the same brief without any model call (offline-safe)
Constraint	The LLM never computes statistics, geometry or physics itself
Disclosure	Record the tool, model name, and key prompts in docs/AI_USE.md

08 DATA ACQUISITION PIPELINE

8.1 NASA FIRMS Integration

Source	Endpoint / Method	Use Case
FIRMS Bulk Archive	Download CSV/SHP from FIRMS archive page	Multi-year historical data for MODIS and VIIRS; do this before the event
FIRMS Area API	GET https://firms.modaps.eosdis.nasa.gov/api/area/csv/{MAP_KEY}/{source}/{bbox}/{days}	Recent days (last 10) to supplement archive; rate-limited to 5000 tx / 10 min
Earthdata Login	OAuth2 for data access	Required for FIRMS MAP_KEY and bulk downloads

8.2 Supporting Data Sources

Source	What	Purpose
NASA GIBS	Web Map Tile Service imagery	Context basemap (true-colour, fire RGB) for the map view
NASA POWER	Surface meteorology (temperature, precipitation, wind)	Weather context for anomaly explanation
NASA MCD64A1	MODIS Burned Area product	Independent cross-check of harmonised seasonal shape (optional)
Local data (optional)	Bangladesh Met Dept records	Partner dataset for NASA-data-usage score

8.3 Caching & Fixture Strategy

- Raw layer: one Parquet file per product per year, unmodified FIRMS columns (git-ignored)
- Demo fixtures: a small subset (~30 days × pilot area) committed to the repo in cache/demo_fixtures/
- GIBS tiles: pre-download for the pilot region + date range; store as PNG/WebP in cache/tiles/
- Acquisition script: `make cache` downloads everything; `make demo` runs with OFFLINE=1
- Retry logic: exponential backoff with 3 retries; on failure, fall back to cached data silently

09 TECHNOLOGY STACK

Layer	Technology	Why
Language	Python >= 3.14	Strongest data science + ML ecosystem; team familiarity
API Framework	FastAPI + uvicorn	Async, auto-docs (OpenAPI/Swagger), orjson serialisation
Data Processing	pandas, numpy, DuckDB	DuckDB gives SQL over Parquet without a server; spatial extension for geo queries
Geospatial	shapely, geopandas, h3-py (optional)	Spatial operations, bounding box filtering, optional H3 hexagonal indexing
ML — Clustering	scikit-learn (DBSCAN)	Haversine metric, no need to specify cluster count, well-tested
ML — Calibration	statsmodels (Poisson GLM), scipy	Standard statistical modelling with robust standard errors
ML — Stretch	LightGBM (gradient boosting)	Fast, handles mixed features, calibrated probabilities via isotonic regression
Validation	pytest, scipy.stats	Unit tests for deterministic functions; statistical tests for step-changes
Serialisation	orjson, Parquet (pyarrow)	Fast JSON for API; columnar Parquet for storage + analytics
Frontend (teammate)	MapLibre GL, D3.js / Observable Plot	Open-source map renderer; powerful charting for calendar heatmap
Optional: LLM	Claude API or OpenAI API	Explanation-only agent with template fallback
Optional: DB	PostgreSQL + PostGIS	Only if DuckDB is insufficient; not needed for hackathon MVP

10 REPOSITORY STRUCTURE

```
firefusion/
├── LICENSE                     # Apache-2.0
├── README.md                   # Project overview, setup, demo instructions
├── Makefile                    # make cache, make pipeline, make demo, make test
├── pyproject.toml              # Dependencies, pinned versions
├── config/
│   └── aoi.yaml               # bbox, dates, cluster radius, quality filter, timezone
├── src/
│   ├── acquire/
│   │   ├── firms.py           # FIRMS bulk + area API download
│   │   ├── gibs.py            # GIBS tile pre-fetcher
│   │   ├── power.py           # NASA POWER weather context
│   │   └── safe.py            # Retry, fallback, offline wrapper
│   ├── compute/
│   │   ├── schema.py          # Canonical schema definition (pydantic)
│   │   ├── normalise.py       # FIRMS → canonical conversion
│   │   ├── cluster.py         # DBSCAN per-day clustering
│   │   ├── calibrate.py       # Detectability matching, Poisson GLM, LightGBM
│   │   ├── baseline.py        # Day-of-year baseline + anomaly percentile
│   │   ├── aggregate.py       # Pre-compute daily + cell aggregates
│   │   └── validate.py        # Overlap correlation, step-change, sensitivity
│   ├── api/
│   │   ├── main.py            # FastAPI app, CORS, startup
│   │   ├── routes/
│   │   │   ├── calendar.py    # /calendar endpoint
│   │   │   ├── cells.py       # /cells endpoint
│   │   │   ├── clusters.py    # /clusters/{id} + /clusters/search
│   │   │   ├── validation.py  # /validation endpoint
│   │   │   ├── anomaly.py     # /anomaly + /baseline endpoints
│   │   │   ├── brief.py       # /brief endpoint (template + optional LLM)
│   │   │   └── meta.py        # /meta + /health endpoints
│   │   └── models.py          # Pydantic response models
│   └── agents/
│       └── explain.py         # Optional LLM explainer (no computation)
├── cache/                      # Parquet files (git-ignored)
│   └── demo_fixtures/          # Small subset for offline demo (committed)
├── tests/
│   ├── test_normalise.py
│   ├── test_cluster.py
│   ├── test_calibrate.py
│   ├── test_validate.py
│   └── test_api.py
├── docs/
│   ├── AI_USE.md              # AI disclosure (required)
│   ├── METHODS.md             # Scientific methodology
│   └── LIMITATIONS.md         # What this does and does not mean
└── web/                        # Static frontend (teammate)
    ├── index.html
    ├── map.js
    └── calendar.js
```

11 TESTING & VALIDATION STRATEGY

Test Category	What	Tool
Unit tests	Deterministic synthetic cases: a fire seen by both sensors counts once; VIIRS-only small fire counts once; cell-boundary cases; date-boundary cases for local time	pytest
Schema tests	Validate that normalised output matches canonical schema; check for nulls, type mismatches, invalid enum values	pytest + pydantic
Statistical tests	Step-change test (segmented regression or CUSUM); overlap correlation; bootstrap confidence intervals	scipy, statsmodels
Sensitivity tests	Run pipeline at multiple cluster radii (0.75, 1.0, 1.5, 2.0, 3.0 km) and report stability	Custom script
API tests	Endpoint contract tests: correct status codes, response shapes, provenance present, edge cases (empty bbox, invalid dates)	pytest + httpx
Integration test	Full pipeline from raw Parquet → API response → validate output matches expected fixtures	make test-integration
Offline test	Run with OFFLINE=1 and verify all endpoints return valid data from fixtures	make test-offline

12 DEPLOYMENT ARCHITECTURE

For a 48-hour hackathon, keep deployment simple. The system is designed to run entirely on a single machine.

Component	Deployment	Notes
Backend (FastAPI)	uvicorn on localhost:8000	Single process is sufficient for demo; production would use gunicorn with multiple workers
Frontend	Static files served by FastAPI or a simple HTTP server	No build step needed for vanilla HTML/JS; if using a framework, pre-build
Database	DuckDB (in-process, no server)	Reads from Parquet files; no installation needed
Data cache	Local filesystem (cache/ directory)	Pre-populated before the event
Optional: Docker	docker-compose.yml with one service	Only if team wants portable demo; not required

Production Architecture (If Advanced to Global)

Post-hackathon scaling path (for reference only, not built during the event):

- Backend: containerised FastAPI on a cloud platform (Fly.io, Railway, or AWS ECS)
- Storage: Parquet on S3 + DuckDB for queries; or PostgreSQL + PostGIS for multi-user access
- Caching: Redis for API response caching
- CI/CD: GitHub Actions for tests + deployment on push to main

13 48-HOUR HACKATHON EXECUTION PLAN

12.1 Key Dates & Deadlines

Date	Milestone	Action
1 Oct 2026	240-second prescreening video due (11:59 PM)	Record concept video from this document; no real results needed yet
28 Oct 2026	Full challenge statements + named datasets published	Re-check product names, AOI, any scope changes
13 Nov 2026	Detailed submission/judging guides; Bangladesh event start	Read submission guide that morning; final data refresh
13–14 Nov (BD) / 14–15 Nov (global)	Hackathon weekend	Execute the plan below

12.2 Pre-Event Preparation (Now to 12 Nov)

25. Register for Earthdata Login; get FIRMS MAP_KEY and NASA API key; test each with one call
26. Create public repository with licence, README skeleton, docs/AI_USE.md, and the layout from Section 10
27. Download multi-year FIRMS archives for MODIS and VIIRS over the AOI; write Parquet; record row counts
28. Implement and unit-test normalisation and grid-collapse baseline; produce first raw vs baseline-harmonised plot
29. Implement clustering; run the radius sensitivity table
30. Implement overlap validation and Poisson sensor factor; freeze metrics that go on screen
31. Cache GIBS tiles, POWER data, and fixtures; build `make cache` and `make demo` with OFFLINE=1
32. Agree API contract (Section 6) with frontend teammate; both work in parallel with mock JSON
33. Pre-record a fallback demo video as safety net

12.3 Day 1 & Day 2 Task Breakdown (Developer Tasks)

Block	Developer Tasks	Deliverable
Day 1 Morning	Re-verify data against 28 Oct statement; rebuild pipeline end-to-end; lock cluster radius and quality filter; publish first API endpoints	Pipeline runs green; API serves /calendar and /cells with real data
Day 1 Afternoon	Serve calendar and cells endpoints; make raw/harmonised toggle work against the API; help frontend integrate	Raw ↔ Harmonised toggle works; first visual on screen
Day 1 Evening	Baseline and anomaly endpoints; validation endpoint; help record local judging video with real numbers	Anomaly + validation endpoints live; judging video recorded
Day 2 Morning	Uncertainty bands; methods/validation panel data; template brief (LLM optional); performance + offline test	/brief endpoint; offline test passes; all "Should" items done
Day 2 Noon	Freeze code; tag release; verify repo opens in private window; complete project page + AI-use disclosure; submit	Tagged release; project page complete; submission confirmed

12.4 Cut Order (If Behind)

Drop features in this order, protecting the core to the end:

34. Drop second demonstration region (keep Bangladesh only)
35. Reduce time range to 5 years instead of full 20+
36. Drop the anomaly engine (baseline + seasonal view)
37. Drop the learned co-detection model (Layer C)
38. Drop the LLM explainer (keep template brief)
39. NEVER drop: harmonisation, validation, or the raw/harmonised toggle

PROTECT Keep one memorable moment: the toggle where the artefact fades and the seasonal signal appears. Protect it above all other features.

14 JUDGING ALIGNMENT STRATEGY

Criterion	Points	How This Design Earns It
Impact (1–20)	Target: 16+	Continuity of fire record as MODIS retires; Bangladesh and neighbouring burning-season framing (crop residue, hill-region shifting cultivation, forest fire) with measured numbers
Creativity (1–20)	Target: 16+	Raw/harmonised toggle that visibly removes a sensor artefact; cluster evidence panel showing which sensor saw what; confidence-aware scoring
Validity (1–20)	Target: 18+	Overlap-period validation, step-change test, bootstrap confidence intervals, deterministic code with tests, methods panel with limitations
Relevance (1–20)	Target: 18+	FIRMS MODIS + VIIRS are the core input, named on screen and on camera; GIBS context; NASA POWER weather data
Presentation (1–20)	Target: 15+	One clear story in 240 seconds: problem → artefact → fix → proof → use
NASA Data Usage (1–5)	Target: 5	FIRMS + NASA POWER + optionally MCD64A1 + a partner/local dataset
Free Points (0/1 × 3)	Target: 3/3	Name the challenge, public repo that opens without login, complete project page

15 RISKS & MITIGATIONS

Risk	Impact	Probability	Mitigation
Harmonised series still shows step at handover	Weakens core claim	Medium	Layers A+B together; show residual honestly with interval; test several radii
Data volume too large for time	Slow clustering, missed demo	Medium	Start with Bangladesh box; per-day parallel clustering; precomputed aggregates

FIRMS access or rate limits	Missing data at event time	Low	Bulk pre-download; fixtures; OFFLINE=1
Sensor start dates or field values differ from assumptions	Schema bugs	Medium	Inspect real files first; schema tests; keep raw layer untouched
Over-claiming (fire spread, evacuation)	Validity penalty, misuse risk	Low	Fixed wording rules; limitations page; no prediction of spread
UI clutter / poor UX	Lower presentation score	Medium	Simple date/region filters, one toggle, clear legend
Official statement changes scope on 28 Oct	Rework	Medium	Config-driven product names and AOI; re-read on release day
LLM invents numbers	Validity penalty	Medium	Numbers only from JSON object; template fallback; cite-check test
Demo wifi fails	Cannot show product	Medium	OFFLINE=1 mode; pre-recorded fallback video
Team member unavailable	Lost capacity	Low	Pre-build as much as possible; documented API contract enables parallel work

16 SUGGESTED PROJECT TOPIC & PITCH

15.1 Three Topic Framings

Three framings of the same technical core; they differ only in the story told to judges. Option A is recommended.

Option	Title & Tagline	Why Choose It
A (Recommended)	FireFusion: One Continuous Fire Record for South Asia — Keeping 25 years of fire history comparable as MODIS retires and VIIRS takes over	Directly matches challenge text and sensor-retirement news; strongest Validity story (artefact removed, proof on screen); simplest to explain in 240 seconds
B	Burn Calendar Bangladesh — When and where the region burns, on one trustworthy calendar	Stronger local Impact and Bangladesh framing (crop residue, hill-region, forest burning); slightly less about the science, so keep validation visible
C	Sensor Handover Watch — An early-warning view that flags when a new sensor's counts differ from the old one's expectation	Creative twist for early warning; needs baseline + anomaly engine working, so higher risk in 48 hours

15.2 Recommended Narrative for Option A (~240 seconds)

Time	Content
0:00–0:30	Team VU_Surveyors, all members named. Fire history matters for responders and land managers; the satellite record is changing under our feet.

0:30–1:15	The problem: MODIS at 1 km and VIIRS at 375 m see fires differently; Suomi-NPP delivery ends 1 Nov 2026 and MODIS is being retired. Stitching creates a false jump.
1:15–2:15	Approach: FIRMS data → common schema → event clustering → detectability matching → sensor factor with interval. Name the datasets on camera.
2:15–3:15	Demo: toggle raw to harmonised over the pilot area; show validation numbers and the calendar; open one cluster to see both sensors' evidence.
3:15–4:00	Impact and honesty: what it means and does not mean; open-source, offline-capable; next steps (more regions, NOAA-21 continuity, MCD64A1 cross-check).

15.3 Hackathon-Level Advice

- Secure the three free points first (challenge named, public repo, project page complete) and keep them safe
- Keep one memorable moment: the toggle where the artefact fades and the seasonal signal appears
- Write docs/AI_USE.md as you go; cite every dataset with an ID and URL
- If the team has any women members, the guide lists a +5% aggregate bonus on the score sheet
- For the prescreening video (concept only), use the narrative above without quoting result numbers you haven't measured yet

Additional Suggested Project Names

Name	Tagline	Fit
FireFusion	Confidence-Aware Wildfire Intelligence	Best fit — fusing observations, not displaying them
FireSync	Harmonizing NASA Fire Observations	Clear and descriptive
ThermalBridge	Bridging MODIS and VIIRS Fire Records	Emphasises the transition problem
PyroFusion	Multi-Sensor Fire Activity Harmonization	More scientific feel
FireLens	NASA Fire Activity Intelligence	Broader scope, less specific

17 OPEN QUESTIONS

40. Final pilot area: Bangladesh only, or Bangladesh plus neighbouring regions? (affects data volume and story)
41. Which record do we harmonise: MODIS-only until 2012 then VIIRS, or the combined MODIS + VIIRS record throughout? Decide and state it in the methods panel.
42. Which partner or local data will we add for the NASA-data-usage score (e.g., Bangladesh Meteorological Department records)?
43. Who owns the frontend, and can we agree the JSON contract this week?
44. Confirm event dates, local judging upload time, and video rules with the Local Lead.
45. Team name consistency: files say "Team Surveyors"; registration says "VU_Surveyors". Use one spelling everywhere.
46. PostGIS vs. offline-first: DuckDB with spatial extension gives both SQL and no server. Keep PostGIS optional.

18 Review of Provided Resources

Resource Summary

Resource	What It Is	How Useful It Is
Team strategy PDF (86 pages)	Guidebook covering all 14 challenges with strategy analysis, team ranking, and "FireFusion" proposal for challenge 9	Useful for scope: recommends 7–30 day pilot window, common schema, UTC normalisation, spatio-temporal clustering, cluster evidence panel
Build guide (HTML)	Technical companion for Bangladesh local events: shared architecture, FIRMS endpoints, 48-hour plan, runnable harmonisation snippet, judging criteria, submission checklist	Most actionable of all files; this design builds on it directly
NASA Space Apps 2026 Guidebook	Overview of all 14 challenges with difficulty ratings, skills/tools, risks, AI agent roles, judging criteria	Confirms Challenge 9 as "MEDIUM" difficulty with strong data availability; validates our scope decisions
ChatGPT analysis	Detailed review of the problem with architecture recommendations, ML features, frontend views, demo script	Good starting point; this document corrects several issues (confidence filtering, sensitivity gap) and adds implementation detail

What Is Strong in the Existing Plans

- Clear success definition: accurate, time-based map reconciling MODIS (longer record) and VIIRS (higher resolution)
- Sound engineering: offline-first demo, deterministic science code, models only explain, every dataset cited
- Good scoping discipline: cut list protects the core (one region, five years, drop anomaly before harmonisation)
- Honest limitations: hotspots are not fire boundaries, confidence fields are not comparable, no fire-spread claims

Issues Identified & Corrected in This Design

#	Issue	Correction
1	Confidence filter not portable across sensors (sample SQL uses confidence >= 50)	Map both to common three-class scale before filtering
2	Grid-collapse alone won't equalise sensitivity	Added detectability matching + Poisson sensor factor (Section 7.3)
3	'sensors agreeing' counts platforms, not families	Use sensor_family column (MODIS/VIIRS), keep platform separately
4	UTC date vs local date	Calendar uses Asia/Dhaka local date; UTC kept for provenance
5	Unverified headline numbers ('counts triple')	Treat as hypothesis; measure on real data before pitching
6	Sensor end-of-life dates need citation	Say 'being retired' rather than one exact date; cite NASA pages
7	Datasets not final until 28 Oct	Config-driven product names and AOI; re-read on release day
8	Event date discrepancies between files	Confirm with Local Lead

9	Team name spelling inconsistency	Use 'VU_Surveyors' everywhere
10	PostGIS vs offline-first tension	DuckDB with spatial extension as default; PostGIS optional

19 APPENDIX: ADDITIONAL CODE SAMPLES

A.1 Configuration File (config/aoi.yaml)

```yaml
# FireFusion configuration
aoi:
  name: "Bangladesh"
  bbox: [88.0, 20.0, 93.0, 27.0]  # [lon_min, lat_min, lon_max, lat_max]
  timezone: "Asia/Dhaka"

data:
  start_date: "2000-11-01"  # Terra MODIS start
  end_date: "2026-09-30"
  products:
    - "MODIS_C6.1"
    - "VIIRS_SNPP_NRT"
    - "VIIRS_NOAA20_NRT"

clustering:
  eps_km: 1.5
  min_samples: 1
  sensitivity_radii: [0.75, 1.0, 1.5, 2.0, 3.0]

quality:
  min_conf_class: 1  # 0=low, 1=nominal, 2=high
  exclude_static: true

calibration:
  overlap_start: "2012-01-01"
  overlap_end: "2026-09-30"
  bootstrap_n: 1000
  random_seed: 42
```

A.2 FastAPI Application Entry Point

```python
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from src.api.routes import calendar, cells, clusters, validation, anomaly, brief, meta

app = FastAPI(
    title="FireFusion API",
    description="Sensor-harmonised wildfire intelligence",
    version="1.0.0"
)

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],  # Restrict in production
    allow_methods=["GET"],
    allow_headers=["*"]
)

app.include_router(calendar.router, prefix="/calendar", tags=["Calendar"])
app.include_router(cells.router, prefix="/cells", tags=["Cells"])
app.include_router(clusters.router, prefix="/clusters", tags=["Clusters"])
app.include_router(validation.router, prefix="/validation", tags=["Validation"])
```

```
app.include_router(anomaly.router, prefix="/anomaly", tags=["Anomaly"])
app.include_router(brief.router, prefix="/brief", tags=["Brief"])
app.include_router(meta.router, prefix="/meta", tags=["Metadata"])

@app.get("/health")
async def health():
    return {"status": "ok", "mode": "offline" if OFFLINE else "live"}
```

A.3 Makefile Commands

```
.PHONY: cache pipeline demo test

cache:                          # Download all data
    python -m src.acquire.firms
    python -m src.acquire.gibs
    python -m src.acquire.power

pipeline:                       # Run full harmonisation pipeline
    python -m src.compute.normalise
    python -m src.compute.cluster
    python -m src.compute.calibrate
    python -m src.compute.baseline
    python -m src.compute.aggregate
    python -m src.compute.validate

demo:                           # Start offline demo
    OFFLINE=1 uvicorn src.api.main:app --port 8000

test:                           # Run all tests
    pytest tests/ -v --tb=short

test-offline:                   # Verify offline mode works
    OFFLINE=1 pytest tests/test_api.py -v
```

A.4 Team Role Split

Role	Person	Owns
Backend + AI/ML (You)	Developer	Acquisition, schema, harmonisation engine, calibration, validation, API, tests, system design
Map / Frontend Developer	Teammate	MapLibre map, calendar heatmap, toggle, evidence panel; consumes mock JSON first
UI/UX Designer	Teammate	Legends, filter simplicity, uncertainty display, methods page
Science & Documentation	Teammate	Sensor facts, citations, METHODS.md, LIMITATIONS.md, project page text
Pitch & Submission	Teammate	Prescreening video, local + 30-second global videos, subtitles, final checklist

Sources Consulted

- NASA FIRMS MODIS active fire description (product coverage; Suomi-NPP delivery ends 1 Nov 2026)
- NASA Earthdata / LAADS DAAC pages on the MODIS to VIIRS transition
- NASA Earthdata page on the VIIRS I-band 375 m active fire product
- NASA Space Apps Challenge 2026 Team Guidebook
- NASA Space Apps Challenge 2026 Bangladesh Build Guide

- All 14 Problem Analysis and Selection (team document)

Field-level details (thresholds, platform labels, start dates) should be checked against the current FIRMS attribute tables before the schema is frozen.